COURSE HANDOUT

Dated: 01.07.2026

Academic Year	: 2026-27
Degree	: B.Tech
School	: School of Advanced Computing
Program	: Computer Science and Engineering
Semester	: 3
Regulation	: AR23 (Autonomous – JNTUGV Regulation)
Course Title	: Object Oriented Programming with JAVA
Course Code	: 23CS306
L-T-P-C Structure	: 3-0-0-3
Instructor(s)	**:** Mr. G. Dharma Raju, Mr. G Suneel, Dr. Radha Rani, Mr. G. Ravi Kumar Mrs. P Madhavi, Mr. G Surendar
Course coordinator	**:** Mr. G. Surendar

1.0 Course description

This course introduces the fundamental concepts of programming from an object-oriented perspective and the necessity to learn the importance of object-oriented programming (OOP) and the difference between the traditional procedure-oriented programming and object-oriented programming. In the structured oriented programming languages, methods define the structure of the programs, they are basic building blocks and data has secondary role while in object-oriented programming languages the data has the principal role. Methods belong to the data, without the data, the method does not have any meaning (Except static methods). Data and methods together make up the object. OOP tries to model the real world.

This course introduces the basic programming in java. Concepts and usage of class, object and constructors are introduced. Java as a class-based and pure OOP language is used to demonstrate and implement appropriate concepts and techniques. The students are exposed to the concepts, fundamental syntax, and the thought processes behind object-oriented programming.

At the end of the course the students will be able to use the concepts of data abstraction, inheritance, polymorphism, exception handling and Threads. The course also introduces the concepts of developing Applet and event handling which enable them to design interactive applications.

2.0 Course Objectives

After completing this course, students will be able to:

1. Understand the fundamentals of object-oriented concepts through java.
2. Learn about classes and objects
3. Understand the purpose of polymorphism, overloading and interfaces
4. Implements application programs using different types of inheritance
5. Create and import packages in developing projects using java
6. Explore concepts of concurrent programming by using multithreading
7. Handle runtime errors through exception handling mechanism
8. Write interactive applications through applets

3.0 Course Outcome

At the end of the course students are able to:

1. Apply the object-oriented programming concepts to develop modular and reusable software Solutions
2. Develop applications using different types of inheritances
3. Implement user-defined packages to promote code reusability and modular programming
4. Analyse and recover runtime exceptions arise in the applications

5. Apply parallel processing applications using threads
6. Develop interactive applications for standalone and Internet

4.0 CO-PO/PSO Mapping

CO	PO1	PO2	PO3	PO12	PSO2
1	3	3	1	2	1
2	3	3	1	2	1
3	3	3	1	2	1
4	2	2	2	2	2
5	3	2	2	2	2
6	3	2	2	2	2

3–Strongly linked | 2–Moderately linked | 1–Weakly linked

5.0 Course Syllabus

Unit I: Introduction to Java **12 Hours**

Overview of Object Oriented Programming principles, Importance of Java to the Internet, Byte code, Methods, classes and instances, Data types, arrays, control statements, simple java program, Classes and Objects– constructors, methods, access control, this keyword, overloading methods and constructors, garbage collection, java.lang – String, StringBuffer, StringBuilder, Type wrappers – Integer, Character, Float, Double, Boolean.

Features of object oriented programming–Java History–Computer Programming Hierarchy–Role of Java Programmer in Industry

Unit II: Inheritance, Packages & Interface **12 Hours**

Inheritance: Basics, Using super, Multilevel Hierarchy, Method overriding, Dynamic Method Dispatch, Using Abstract classes, Using final with inheritance, Sealed Classes.

Java Collections – Arrays, ArrayList, Hashtable, StringTokenizer, Vector.

Interface: Defining an interface, differences between classes and interfaces, implementing interface, variables in interface and extending interfaces, Default Methods in Interfaces, Static Methods in Interface

Packages: Defining, Creating and Accessing a Package, Understanding CLASSPATH, importing packages, Member access rules.

Nested–Inner Class & Anonymous Classes–Generic Class Types

Unit III: Exception Handling & Multithreading **11 Hours**

Exception handling: Concepts and benefits of exception handling, exception hierarchy, usage of try, catch, throw, throws and finally, built-in and User Defined Exceptions. Multithreading: Definition thread, thread life cycle, creating threads, synchronizing threads, daemon threads. *Control Flow in Exceptions– JVM reaction to Exceptions– Inter Communication of Threads– Critical Factor in Thread–Deadlock*

Unit IV: Applets & Event Handling **13 Hours**

Applets: Concepts of Applets, life cycle of an applet, types of applets, creating applets, passing parameters to applets, The AWT class hierarchy, user interface components labels, button, Text components.

Event Handling: Events, Delegation event model, handling mouse and keyboard events, Adapter classes, inner classes, Compare basic AWT components with swing components, More user interface components-canvas, scrollbars, check box, choices, lists panels–scroll pane, dialogs, menu bar, layout managers.

Applet Security Policies–Anonymous Inner classes a Short–cut to Event Handling–Java Utilities (java.util Package) **Total: 48 Hours**

6.0 Textbooks, References and Web Resources

Textbooks

1. H. Schildt, Java: The complete reference, 7th Ed., TMH, 2016.
2. T. A. Budd, An Introduction to Object-Oriented Programming, 3rd Ed., Addison Wesley Longman, 2012.

Reference Books

1. Dietal & Dietal, Java: How to Program, 8th Ed., PHI, 2010.

2. E. Balaguruswamy, Programming with Java A Primer, 4th Ed., Tata McGraw Hill Companies, 2009.
3. C. S. Horstmann and G. Cornell, Core Java, Vol 1. Fundamentals, 7th Ed., Pearson Education, 2014.
4. C. Horstmann, BIG JAVA Compatible with Java 5 & 6, 3 rd Ed., Wiley Publishers, 2008.

7.0 Detailed Session Plan

Session No.	Unit	CO	Topic	Teaching Method	Reference
1	1	1	Overview of Object-Oriented Programming principles	PPT + Chalk & Talk	T1,R1
2	1	1	Importance of Java to the Internet, Byte code,	PPT + Chalk & Talk	T1,R1
3	1	1	Methods, classes and instances,	PPT + Chalk & Talk	T1,R1
4	1	1	Data types, Arrays	PPT + Chalk & Talk	T1,R1
5	1	1	control statements,	PPT + Chalk & Talk	T1,R1
6	1	1	simple java program,	PPT + Chalk & Talk	T1,R1
7	1	1	Classes and Objects	PPT + Chalk & Talk	T1,R1
8	1	1	Constructors, methods	PPT + Chalk & Talk	T1,R1
9	1	1	access control, this keyword,	PPT + Chalk & Talk	T1,R1
10	1	1	overloading methods, constructors	PPT + Chalk & Talk	T1,R1
11	1	1	garbage collection, java.lang-String, StringBuffer, StringBuilder.	PPT + Chalk & Talk	T1,R1
12	1	1	Type wrappers Integer, Character, Float, Double, Boolean	PPT + Chalk & Talk	T1,R1
13	2	2	Inheritance: Basics,	PPT + Chalk & Talk	T1,R1
14	2	2	Using super, Multilevel Hierarchy	PPT + Chalk & Talk	T1,R1
15	2	2	Method overriding, Dynamic Method Dispatch,	PPT + Chalk & Talk	T1,R1
16	2	2	Using Abstract Classes	PPT + Chalk & Talk	T1,R1
17	2	2	Using final with inheritance, sealed classes	PPT + Chalk & Talk	T1,R1
18	2	2	Interface: Defining an interface, differences between classes and interfaces	PPT + Chalk & Talk	T1,R1
19	2	2	implementing interface, variables in interface and extending interfaces	PPT + Chalk & Talk	T1,R1
20	2	2	Default Methods in Interfaces, Static Methods in Interface	PPT + Chalk & Talk	T1,R1
21	2	3	Java Collections- Arrays, ArrayList, Hashtable	PPT + Chalk & Talk	T1,R1
22	2	3	StringTokenizer, Vector	PPT + Chalk & Talk	T1,R1
23	2	3	Packages: Defining, Creating and Accessing a Package,	PPT + Chalk & Talk	T1,R1

			Understanding CLASSPATH		
24	**2**	**3**	importing packages, Member access rules	**PPT + Chalk & Talk**	**T1,R1**
25	**3**	**4**	Exception handling: Concepts and benefits of exception handling.	**PPT + Chalk & Talk**	**T1,R1**
26	**3**	**4**	exception hierarchy	**PPT + Chalk & Talk**	**T1,R1**
27	**3**	**4**	usage of try, catch,	**PPT + Chalk & Talk**	**T1,R1**
28	**3**	**4**	Throw	**PPT + Chalk & Talk**	**T1,R1**
29	**3**	**4**	throws and finally,	**PPT + Chalk & Talk**	**T1,R1**
30	**3**	**4**	built-in Exceptions	**PPT + Chalk & Talk**	**T1,R1**
31	**3**	**4**	User Defined Exceptions.	**PPT + Chalk & Talk**	**T1,R1**
32	**3**	**5**	Multithreading: Definition thread	**PPT + Chalk & Talk**	**T1,R1**
33	**3**	**5**	thread life cycle,	**PPT + Chalk & Talk**	**T1,R1**
34	**3**	**5**	creating threads, synchronizing threads	**PPT + Chalk & Talk**	**T1,R1**
35	**3**	**5**	daemon threads	**PPT + Chalk & Talk**	**T1,R1**
36	**4**	**6**	Applets: Concepts of Applets, differences between applets and applications	**PPT + Chalk & Talk**	**T1,R1**
37	**4**	**6**	life cycle of an applet, types of applets, creating applets, passing parameters to applets,	**PPT + Chalk & Talk**	**T1,R1**
38	**4**	**6**	The AWT class hierarchy, user interface components labels	**PPT + Chalk & Talk**	**T1,R1**
39	**4**	**6**	button and Text components	**PPT + Chalk & Talk**	**T1,R1**
40	**4**	**6**	Event Handling: Events, Delegation event model,	**PPT + Chalk & Talk**	**T1,R1**
41	**4**	**6**	handling mouse and keyboard events	**PPT + Chalk & Talk**	**T1,R1**
42	**4**	**6**	Adapter classes, inner classes,	**PPT + Chalk & Talk**	**T1,R1**
43	**4**	**6**	Compare basic AWT components with swing components,	**PPT + Chalk & Talk**	**T1,R1**
44	**4**	**6**	user interface components-canvas	**PPT + Chalk & Talk**	**T1,R1**
45	**4**	**6**	More user interface components- scrollbars	**PPT + Chalk & Talk**	**T1,R1**
46	**4**	**6**	check box, choices, lists	**PPT + Chalk & Talk**	**T1,R1**
47	**4**	**6**	Panels-scroll pane, dialogs, menu bar, layout managers.	**PPT + Chalk & Talk**	**T1,R1**
48	**4**	**6**	Layout Managers	**PPT + Chalk & Talk**	**T1,R1**

8.0 Assessment Scheme

Component	Duration (minutes)	Marks	% of weightage	Date & Time	Venue
Continuous Assessment – 1	90	40	20 (80% of highest marks plus 20% of lowest marks)	02.09.2026 To 04.09.2026	Block-2, 5, 6
Continuous Assessment – 2	90	40		04.11.2026 To 10.11.2026	Block-2, 5, 6
Open Book/Assignment Examination	120	30	10	04.11.2026 To 10.11.2026	Block- 2, 5, 6
End Semester Examination	180	70	70	17.11.2026 To 27.11.2026	Block- 1, 2, 3,4, 5, 6

Chamber Consultation Hour : Every Friday 4.00 pm to 5.00 pm
Venue : Faculty Cabin
Notices : Department noticeboard.

Signatures of Instructors

Mr. G Dharma Raju - CSE
Mr. G Suneel - CSE
Dr. Radha Rani - CSE
Mr. G Ravi Kumar - CSE
Mrs. P Madhavi – AI&ML

Signature of course coordinator
Mr. G Surendar - AI & DS

Signature of HoD